What was measured?

Laboratory

Controlled driving cycle

A and B: FTP-75 tests at CARB
Same prescribed speed sequence

Real roads

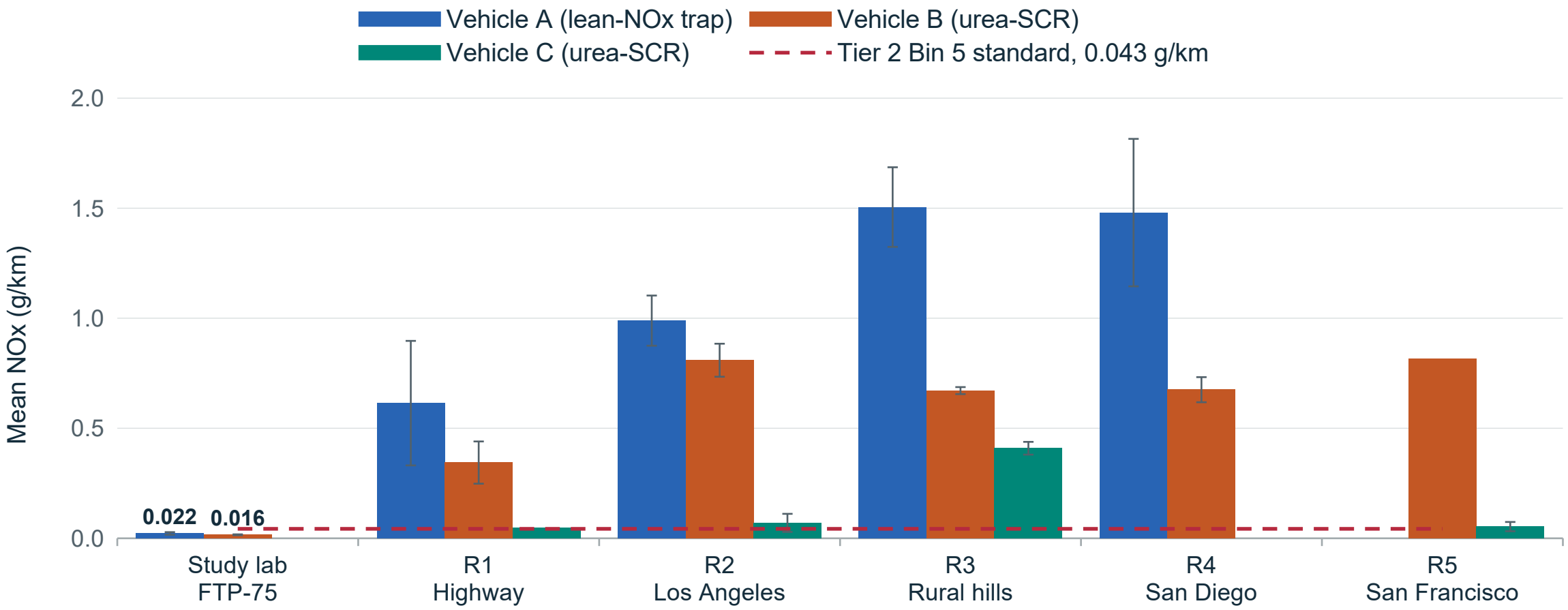
Portable exhaust measurement

Three vehicles, selected California routes
Traffic, hills and engine load vary

NOx = nitrogen oxides g/km = grams per kilometre
U.S. Tier 2 Bin 5 NOx standard: 0.043 g/km at full useful life

WVU/ICCT (2014), pp. 1, 9–10; Table 3.15 p. 57; Table 4.1 p. 59; p. 64. Standard = 0.07 g/mi (Table 2.2 p. 5).

What did the data show?



Bars: route or test means; thin bars: ±1 SD as published (none for single runs). No bar = no test (C lab, C R4, A R5), not zero.

The standard is a certification benchmark, used here as a reference; road routes are not certification tests. Three vehicles: no fleet-wide estimate.

WVU/ICCT (2014), Table 4.3 p. 65; laboratory results p. 64; standard Table 4.1 p. 59. Vehicle identities: EPA NOV (2015), p. 4.

What did the discrepancy mean?

2014

Anomaly

Independent road measurements justified further investigation

2015

Mechanism

EPA described test detection and reduced control on the road

2017

Admitted conduct

Volkswagen admitted facts about design and concealment

The road data showed the gap. It did not, by itself, prove intent.

WVU/ICCT (2014), p. iv, pp. 62–64; EPA NOV (2015), pp. 3–4; VW plea (2017), Exh. 2 pp. 12–14, 21–24, ¶¶ 33–36, 54–55, 63.

Primary sources

2014 research

WVU / ICCT report

15 May 2014
Study design: pp. 1, 9–10, 57
Standard: Table 4.1 p. 59
Results: p. 64; Table 4.3 p. 65

2015 regulator findings

EPA Notice of Violation

18 September 2015
Switch and calibrations: pp. 3–4
WVU vehicles identified: p. 4

2017 admissions

Volkswagen plea agreement

11 January 2017, Exhibit 2
¶¶ 30, 33–36, 50–55, 58, 62–64, 72
Penalties: p. 13

Complete source details and links are also in the accompanying claim register.